What your overcut defence asks for

Measured degradation rates across 73 Formula 1 circuit-compound slopes, 2022–2026

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The quantity

Your D2 is efficient only where a driver can build enough tyre advantage to pass, and your inequality fixes how much: $a = (o + \varepsilon) / \theta$. Section 2.6 takes $\theta = 0.1$ s/lap, $o = 0.55$, $\varepsilon = 0.1$, giving $a = 6.5$ laps on a 40-lap race, so D2 is open to the defender for most of it.

I fit θ per circuit and compound on five seasons of public timing. Of the 57 slopes whose cluster-robust interval excludes zero, 79% degrade more slowly than your 0.1. Your example is close to the most degrading circuit I measure, not a typical one.

What $a = (o + \varepsilon) / \theta$ asks for elsewhere

o (s)	a: lower quartile – median – upper	a > race distance	a > half-distance
0.30	4 – 5 – 8	16 of 73	16 of 73
0.55	7 – 10 – 15	16 of 73	19 of 73
0.80	10 – 15 – 22	18 of 73	25 of 73

At your own $o = 0.55$: a median of 10 laps against your 6.5, quartiles 7 to 15. D2 is unreachable on 16 of 73 circuit-compounds and opens only past half-distance on 19.

I set $\varepsilon = 0$ throughout rather than your 0.1. That is the equal-pace case and the one most favourable to the defence, so every figure above understates the difficulty.

Where it is hardest

Jeddah medium (44 laps of 50), Baku medium (42 laps of 51), Imola medium (39 laps of 63)

Three things this is not

It is not a calibration of o. I hold no overtaking threshold in seconds — what I measure per circuit is an adjacent-pair swap probability per lap, a different object that does not convert into one. So o is yours throughout, varied around your value.

It is not a complete picture. On 16 of the 73 slopes the interval contains zero, so θ is not identified and a is undefined rather than large. I count those as unreachable and exclude them from every average; dividing by them gives Singapore's soft tyre 4,399 laps, out of a slope of 0.0001 s/lap, which measures my noise and not your model.

It is not a judgement on your proofs. They concern a model and hold inside it. What this measures is how much of that model real parameter values leave switched on — and my reading is that D1 and D2 are not equally robust to small θ, since D1 stays well defined as θ falls while D2 divides by it.

What I would value your view on

Whether a linear θ is the right object for D2 at all. If the advantage that lets a car pass is concentrated in a fresh tyre's first laps rather than accumulated linearly, then $a = (o + \varepsilon) / \theta$ understates D2's availability on exactly the low-θ circuits where I report it as unreachable.